

1-1 Measurement and Uncertainty



Source / 來源

- 選修物理(一)課堂講義 (4623R52) 教用版，第1章 不確定度，§ 1-1 測量與不確定度，p. 6
- Halliday, Resnick & Walker, *Fundamentals of Physics*, Ch 1: Measurement
- Young & Freedman, *University Physics with Modern Physics*, Ch 1: Units, Physical Quantities, and Vectors

Examples

Section Scope

Section 1-1 introduces the meaning of error, uncertainty, and standard deviation. The first numbered worked example in the source appears in § 1-2 and is reproduced in the § 1-2 worksheet; no numbered example is printed in § 1-1 itself.

Error and Uncertainty

An **error** is the difference between a measured value and the true value:

$$\text{error} = \text{measured value} - \text{true value}$$

For well-established constants such as the gravitational acceleration or speed of light, an internationally accepted value with very small uncertainty may be used as a practical true value when calculating experimental error.

For most physical quantities—such as height, mass, and temperature—the true value is unknown. Even though infinitely many trials would approach the most probable value of a normal distribution, infinitely many measurements cannot be performed. **Uncertainty** therefore replaces error as a quantitative description of the dispersion or lack of certainty in a measurement. A smaller uncertainty means better measurement quality and a more trustworthy result.

Possible sources of uncertainty include the object being measured, the instrument, the observer, and the environment. The later sections classify these contributions as Type-A and Type-B evaluations.

Population and Sample Standard Deviation

For a population of N values with mean μ ,

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2}, \quad \mu = \frac{1}{N} \sum_{i=1}^N x_i$$

In a real experiment, only a finite sample of n values is available, with $n < N$. Its sample standard deviation is

$$SD = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}, \quad \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Increasing the number of measurements generally makes the statistical estimate more representative of the underlying population.

Glossary

English term	中文	Definition / Note
measurement	測量	Experimental determination of a physical quantity
measured value	測量值	Value obtained from an observation
true value	真值	Ideal value, usually unknown in practice
accepted value	公認值	Reference value accepted for practical comparison
error	誤差	Measured value minus true value
uncertainty	不確定度	Quantified dispersion or lack of certainty
population	母體	Complete set of possible measurements
sample	樣本	Finite subset actually measured
mean	平均值	Arithmetic average $\bar{x}$ or population mean μ
standard deviation	標準差	Measure of data spread
normal distribution	常態分布	Distribution whose central region is most probable